Part A — Concept Application (40%)

1. Data Loading & Exploration

```
import pandas as pd

df = pd.read_csv('suv_data.csv')
print(df.head())
print(df.shape)
print(df.columns.tolist())
print(df.dtypes)
print(df.isnull().sum())
```

This gives you a quick overview of what you're working with — shape confirms row/column count, dtypes catch any parsing issues, and isnull catches gaps before preprocessing.

2. Data Preprocessing

```
from sklearn.preprocessing import LabelEncoder

df.dropna(inplace=True)

le = LabelEncoder()
df['Gender'] = le.fit_transform(df['Gender'])  # Male=1, Female=0

X = df[['Age', 'EstimatedSalary']]
y = df['Purchased']
```

Only Age and EstimatedSalary are used as features, as User ID is just an identifier and doesn't carry predictive value.

3. Train-Test Split

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
print(X_train.shape, X_test.shape)
```

4. Feature Scaling

```
from sklearn.preprocessing import StandardScaler

sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)  # fit only on train, transform test
```

Scaling is fit only on training data to avoid data leakage — applying the same scaler to test keeps the transformation honest.

5. Model Training

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
model.fit(X_train, y_train)
```